



GridWatch AI

Power Grid Failure Advisor

Real-time multi-zone asset failure risk detection
and automated dispatcher triage

SCADA Online

AI Risk Engine

IBM Hackathon

THE CHALLENGE

Aging grids. Invisible risks. Costly failures.



Reactive, not predictive

Most utilities discover transformer failures only after blackouts, equipment damage, and emergency overtime have already hit.



Slow, manual triage

Dispatchers juggle SCADA alarms, weather feeds, and maintenance logs. Critical assets get buried in noise.



Fragmented systems

Health indices, risk factors, and crew schedules live in separate tools. No single source of truth for action.

THE SOLUTION

One AI command center for the entire grid.

GridWatch continuously monitors transformer health across all sectors, scores failure risk in real time using SCADA telemetry + weather + maintenance history, and generates prioritized action plans with automated crew dispatch recommendations.

20+

Live assets

5 sectors

<2h

Critical window

Rapid response

AI

Risk engine

Always scoring



Fleet Registry

Health index • Failure probability • Primary risk factor



Geospatial Risk Map

Live risk topology colored by severity across sectors



Action Plan

Prioritized work orders + field crew dispatch schedule



AI Risk Engine

Continuous scoring, triage automation & BLUF intelligence

Four views. One operational truth.



Dashboard

- ▶ Live risk distribution donut
- ▶ Critical alert banner with BLUF
- ▶ Monitored / Critical / Optimal counts



Fleet Registry

- ▶ Health Index + Failure Probability
- ▶ Primary risk factor per asset
- ▶ Sector filters & inspect actions



Risk Map

- ▶ Geospatial topology of 20 assets
- ▶ Color-coded by live severity
- ▶ Active outage alerts on map



Action Plan

- ▶ Critical & medium priority queues
- ▶ Field crew dispatch schedule
- ▶ Parts lists, duration & ETD

From telemetry to action in seconds.

01

Ingest

SCADA telemetry, weather feeds, maintenance history, and asset metadata stream continuously across all 20 nodes.

02

Score

AI computes Asset Health Index (AHI) and failure probability. Risk drivers (thermal, mechanical, weather) are ranked.

03

Triage

Critical assets trigger immediate directives. Medium risks enter 24h diagnostic queues. Low risks stay baseline.

04

Dispatch

Crew recommendations, parts lists, duration, and ETD are generated and synchronized with active maintenance orders.

What operators see in the first 60 seconds.

3

Critical

Immediate action

1

Medium

Targeted surveillance

16

Optimal

Baseline operation

20

Fleet

5 sectors covered

CRITICAL ALERT — TX-114

Thermal Runaway • North Sector

Top oil temperature exceeded critical threshold under 96% load. Asset Health Index at 23.25%. Failure probability 76.8%. Emergency crew CREW-RAPID-01 dispatched.

ACTIONABLE OUTPUT

Immediate de-energization or emergency load shedding. Primary rapid-response crew within 2 hours.

Parts pre-listed • Duration calculated • ETD set

Fewer blackouts. Faster crews. Safer assets.



Prevent failures before they cascade

Catch thermal runaway, mechanical core vibration, and weather-driven risk while there is still time to act — not after the outage.



Cut triage time dramatically

Operators move from scattered SCADA alarms to a single prioritized action plan with clear deadlines and ranked severity.



Right crew, right place, right parts

Dispatch recommendations include target asset, required parts, duration, and ETD — ready for field execution.



One shared picture of the fleet

Geospatial risk topology and health scoring give leadership and field teams the same real-time view of every sector.



GridWatch AI

See the risk. Act before the failure.

Real-time

SCADA-synced telemetry
across every sector

Intelligent

AI risk scoring &
automated triage

Actionable

Crew-ready dispatch
plans with parts & ETD

Thank you • Questions welcome

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